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Research Interests

- My research investigates **the mechanisms underlying reasoning in large language models**: how reasoning behaviour is induced, whether chain-of-thought reflects the computation that actually drives a model's answer, and how far reasoning skills learned in one domain transfer to another. I approach this by combining **behavioural and mechanistic analysis** with **neuro-symbolic methods**.

This builds on a broader background in **information extraction and language understanding**: event relation extraction, event graph generation, and question answering, and is grounded in applied work with legal and law-enforcement partners, where reasoning failures carry serious consequences.

Keywords: reasoning in LLMs; chain-of-thought faithfulness; neuro-symbolic AI; process supervision; generalisation under distribution shift; event-centric information extraction; evaluation of LLMs.

Employment History

- 2024 – present **Postdoctoral Researcher** University of Sheffield, School of Computer Science.
- 2022 – 2024 **Research Assistant** University of Warwick, Department of Computer Science.
- 2020 – 2024 **Graduate Teaching Assistant** University of Warwick, Department of Computer Science.

Education

- 2020 – 2024 **Ph.D. in Computer Science** Department of Computer Science, University of Warwick (Fully funded by Warwick)
Thesis title: *Understanding Event Relation in Text via Representation Learning and LLM-based Graph Generation*.
- 2017 – 2020 **M.Sc. in Software Engineering** School of Software Engineering, South China University of Technology (Guangzhou, China)
GPA: 3.69/4.00, Thesis title: *A Study of Aspect Embedding and Conflict Sentiment Recognition on Aspect-Based Sentiment Analysis*.
- 2013 – 2017 **B.Sc. in Logic** Sun Yat-Sen University (Guangzhou, China)
GPA: 3.7/4.0, Thesis title: *An Aspect-Based Sentiment Analysis System for Online Car Reviews*.

Selected Publications

- 1 M. Cao, **X. Tan**, et al., “Fundamental reasoning paradigms induce out-of-domain generalization in language models,” in *Findings of the Association for Computational Linguistics: ACL 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds., San Diego, California, United States: Association for Computational Linguistics, Jul. 2026, pp. 33 025–33 046, ISBN: 979-8-89176-395-1. [DOI](#): 10.18653/v1/2026.findings-acl.1653
- 2 S. Lewis-Lim, **X. Tan**, Z. Zhao, and N. Aletras, “Can confidence estimates decide when chain-of-thought is necessary for llms?” In *(To Appear) Findings of the Association for Computational Linguistics: AACL-IJCNLP 2026*, Association for Computational Linguistics, 2026. [URL](#): <https://arxiv.org/abs/2510.21007>

- 3 **X. Tan**, M. Valentino, et al., “Compliance vs. sensibility: On the reasoning controllability in large language models,” in *(To Appear) Findings of the Association for Computational Linguistics: EMNLP 2026*, Association for Computational Linguistics, 2026.  URL: <https://arxiv.org/abs/2604.27251>
- 4 J. Kelsall, **X. Tan**, et al., “A rapid evidence review of evaluation techniques for large language models in legal use cases: Trends, gaps, and recommendations for future research,” *AI & SOCIETY*, pp. 1–19, 2025.
- 5 **X. Tan**, M. Valentino, M. E. Akhter, M. Liakata, and N. Aletras, “Enhancing logical reasoning in language models via symbolically-guided Monte Carlo process supervision,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 31 874–31 888, ISBN: 979-8-89176-332-6.  DOI: 10.18653/v1/2025.emnlp-main.1624
- 6 **X. Tan**, Y. Zhou, G. Pergola, and Y. He, “Cascading large language models for salient event graph generation,” in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 2223–2245, ISBN: 979-8-89176-189-6.  URL: <https://aclanthology.org/2025.naacl-long.112/>

Research Projects and Funding

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| 2024 – present |  Addressing Socio-technical Limitations of LLMs for Medical and Social Computing (AdSoLve)
<i>Funded by UKRI via Responsible AI UK as a Keystone Project.</i> Role: Postdoctoral Researcher.
Research on LLMs and complex reasoning (logical and legal reasoning), developing neuro-symbolic methods for real-world problems. Engage directly with law firms to build an evaluation platform that accurately reflects practitioner requirements. |
| 2023 – 2024 |  NLP to Detect Threatening and Abusive Language Toward Victims
<i>Funded by the OPCS Police STAR Fund (project page).</i> Role: Lead Engineer (team of 3).
Developed a platform that helps police gather evidence from victims’ devices, analysing text conversations with LLMs and providing in-depth, interactive feedback. Generalised and published as a system demonstration at NAACL 2025. |
| 2022 – 2023 |  An Event-Centric Dialogue System for Second Language Learners
<i>Funded by the Warwick RDF Science Development Award.</i>
Built an annotation platform and recruited participants to annotate interestingness in English learning conversations. Curated the first large-scale dataset annotated for interestingness and expected interestingness in teacher–student interactions: IntrEx. |

Teaching and Supervision

Lecturing

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| 2024 – present |  Guest Lecturer , University of Sheffield — <i>Natural Language Processing</i> .
Lectures delivered: <i>Neural Sequence-to-Sequence Models for Machine Translation</i> , <i>In-Context Learning and Retrieval Augmented Generation</i> . |
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Teaching Assistance

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| 2020 – 2024 |  Teaching Assistant , University of Warwick, Department of Computer Science.
<i>Natural Language Processing (M.Sc., 2020–2024); Logic and Verification (B.Sc., 2020–2023); Formal Systems Development (M.Sc., 2021–2022).</i>
Duties: delivering seminars and lectures, advising laboratory experiments, and marking. |
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Teaching and Supervision (continued)

Doctoral Supervision

- Co-supervised **6 Ph.D. students** in total, resulting in **5 peer-reviewed publications** (see Research Publications).

Academic Service

Senior Reviewing Roles

- 2025 – 2026 **Area Chair**, ACL Rolling Review (ACL ARR) — rounds serving *EMNLP 2025* and *EACL 2026*.

Reviewing

- Reviewer, *IEEE Transactions on Audio, Speech and Language Processing*.
- 2023 – 2026 Reviewer, ACL Rolling Review (ACL ARR) — rounds serving *NAACL 2024*, *ACL 2024*, *ACL 2026*, *EMNLP 2026*.
- 2023 Reviewer, *Empirical Methods in Natural Language Processing* (EMNLP).
- Reviewer, *European Chapter of the Association for Computational Linguistics* (EACL).

Programme Committees and Organisation

- 2026 Programme Committee, *4th Workshop on Mathematical Natural Language Processing* (MathNLP) at EMNLP 2026.
- 2025 Programme Committee, *3rd Workshop on Mathematical Natural Language Processing* (MathNLP) at EMNLP 2025.
- 2024 – present Organiser, NLP seminar series, University of Sheffield.

Mentoring

- 2025 Mentor for **2 Student Research Workshop papers** at EACL 2025.

Research Publications

Journal Articles

- 1 V. A. Permadi, **X. Tan**, N. S. Moosavi, and N. Aletras, “No shortcuts to culture: Indonesian multi-hop question answering for complex cultural understanding,” *Transactions of the Association for Computational Linguistics*, vol. 14, pp. 1243–1265, Jul. 2026, ISSN: 2307-387X.  DOI: 10.1162/TACL.a.726
- 2 J. Kelsall, **X. Tan**, et al., “A rapid evidence review of evaluation techniques for large language models in legal use cases: Trends, gaps, and recommendations for future research,” *AI & SOCIETY*, pp. 1–19, 2025.
- 3 H. Ren, Z. Li, Y. Cai, **X. Tan**, and X. Wu, “Learning refined features for open-world text classification with class description and commonsense knowledge,” *World Wide Web*, vol. 26, no. 2, pp. 637–660, 2023.
- 4 X. Zhang, J. Xu, Y. Cai, **X. Tan**, and C. Zhu, “Detecting dependency-related sentiment features for aspect-level sentiment classification,” *IEEE Transactions on Affective Computing*, vol. 14, pp. 196–210, 2023.  URL: <https://api.semanticscholar.org/CorpusID:234308817>
- 5 **X. Tan**, Y. Cai, J. Xu, H.-f. Leung, W.-h. Chen, and Q. Li, “Improving aspect-based sentiment analysis via aligning aspect embedding,” *Neurocomputing*, vol. 383, pp. 336–347, 2020.  URL: <https://api.semanticscholar.org/CorpusID:211158105>

Conference Proceedings

- 1 M. Cao, **X. Tan**, et al., “Fundamental reasoning paradigms induce out-of-domain generalization in language models,” in *Findings of the Association for Computational Linguistics: ACL 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds., San Diego, California, United States: Association for Computational Linguistics, Jul. 2026, pp. 33 025–33 046, ISBN: 979-8-89176-395-1.  DOI: 10.18653/v1/2026.findings-acl.1653
- 2 C. Karouzou, **X. Tan**, and N. Aletras, “An empirical study on preference tuning generalization and diversity under domain shift,” in *(To Appear) Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, 2026.  URL: <https://arxiv.org/abs/2601.05882>
- 3 S. Lewis-Lim, **X. Tan**, Z. Zhao, and N. Aletras, “Can confidence estimates decide when chain-of-thought is necessary for llms?” In *(To Appear) Findings of the Association for Computational Linguistics: AACL-IJCNLP 2026*, Association for Computational Linguistics, 2026.  URL: <https://arxiv.org/abs/2510.21007>
- 4 **X. Tan**, M. Valentino, et al., “Compliance vs. sensibility: On the reasoning controllability in large language models,” in *(To Appear) Findings of the Association for Computational Linguistics: EMNLP 2026*, Association for Computational Linguistics, 2026.  URL: <https://arxiv.org/abs/2604.27251>
- 5 A. Alajrami, **X. Tan**, and N. Aletras, “Fine-tuning on noisy instructions: Effects on generalization and performance,” in *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, K. Inui et al., Eds., Mumbai, India: The Asian Federation of Natural Language Processing and The Association for Computational Linguistics, Dec. 2025, pp. 728–742, ISBN: 979-8-89176-298-5.  URL: <https://aclanthology.org/2025.ijcnlp-long.41/>
- 6 S. Lewis-Lim, **X. Tan**, Z. Zhao, and N. Aletras, “Analysing chain of thought dynamics: Active guidance or unfaithful post-hoc rationalisation?” In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 29 826–29 841, ISBN: 979-8-89176-332-6.  DOI: 10.18653/v1/2025.emnlp-main.1516
- 7 M. Parvatham, **X. Tan**, G. Pergola, and C. Gambia, “What makes a conversation interesting? linguistic features predictive of interest in educational conversations between teachers and learners of english,” in *Proceedings of the Annual Meeting of the Cognitive Science Society*, vol. 47, 2025.
- 8 **X. Tan**, M. Parvatham, C. Gambi, and G. Pergola, “IntrEx: A dataset for modeling engagement in educational conversations,” in *Findings of the Association for Computational Linguistics: EMNLP 2025*, Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 21 830–21 845, ISBN: 979-8-89176-335-7.  DOI: 10.18653/v1/2025.findings-emnlp.1191
- 9 **X. Tan**, M. Valentino, M. E. Akhter, M. Liakata, and N. Aletras, “Enhancing logical reasoning in language models via symbolically-guided Monte Carlo process supervision,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, Suzhou, China: Association for Computational Linguistics, Nov. 2025, pp. 31 874–31 888, ISBN: 979-8-89176-332-6.  DOI: 10.18653/v1/2025.emnlp-main.1624
- 10 **X. Tan**, Y. Zhou, G. Pergola, and Y. He, “Cascading large language models for salient event graph generation,” in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 2223–2245, ISBN: 979-8-89176-189-6.  URL: <https://aclanthology.org/2025.naacl-long.112/>
- 11 **X. Tan** et al., “SafeSpeech: A comprehensive and interactive tool for analysing sexist and abusive language in conversations,” in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, Albuquerque, New Mexico: Association for Computational Linguistics, Apr. 2025, pp. 361–382, ISBN: 979-8-89176-191-9.  URL: <https://aclanthology.org/2025.naacl-demo.31/>

- 12 **X. Tan**, Y. Zhou, G. Pergola, and Y. He, “Set-aligning framework for auto-regressive event temporal graph generation,” in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, Mexico City, Mexico: Association for Computational Linguistics, Jun. 2024, pp. 3872–3892.  URL: <https://aclanthology.org/2024.naacl-long.214>
- 13 **X. Tan**, G. Pergola, and Y. He, “Event temporal relation extraction with Bayesian translational model,” in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, Association for Computational Linguistics, May 2023.  DOI: 10.18653/v1/2023.eacl-main.80
- 14 J. Lu, **X. Tan**, G. Pergola, L. Gui, and Y. He, “Event-centric question answering via contrastive learning and invertible event transformation,” in *Findings of the Association for Computational Linguistics: EMNLP 2022*, Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022.  DOI: 10.18653/v1/2022.findings-emnlp.176
- 15 **X. Tan**, G. Pergola, and Y. He, “Extracting event temporal relations via hyperbolic geometry,” in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, Association for Computational Linguistics, Nov. 2021.  DOI: 10.18653/v1/2021.emnlp-main.636
- 16 **X. Tan**, Y. Cai, and C. Zhu, “Recognizing conflict opinions in aspect-level sentiment classification with dual attention networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, Association for Computational Linguistics, Nov. 2019.  DOI: 10.18653/v1/D19-1342
- 17 K. Yang, Y. Cai, Z. Cai, **X. Tan**, et al., “A new samples selecting method based on k nearest neighbors,” in *2017 IEEE International Conference on Big Data and Smart Computing (BigComp)*, 2017, pp. 457–462.  URL: <https://api.semanticscholar.org/CorpusID:16318950>

References

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